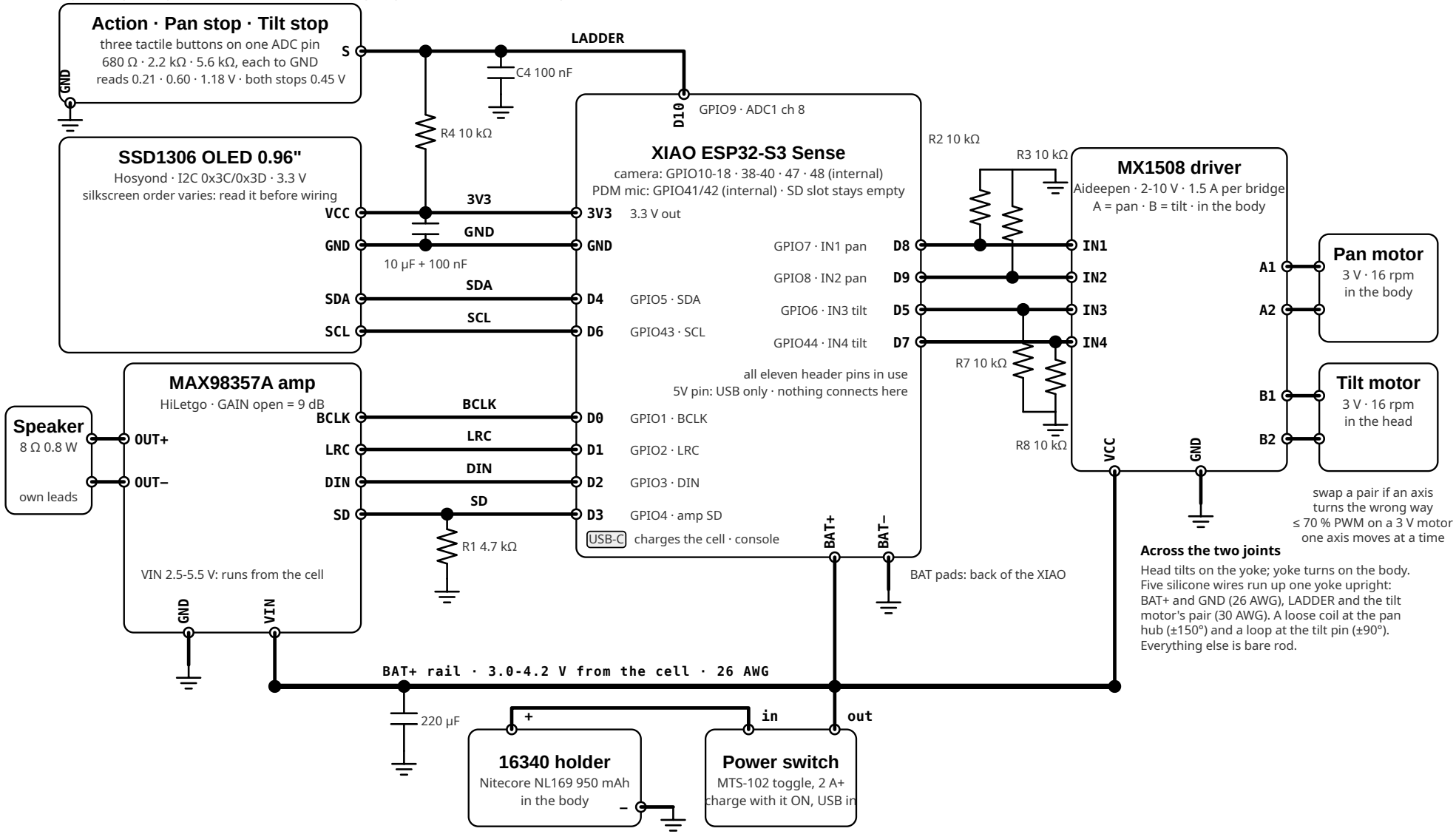
Pocket AI with camera, Rev B: every connection on one sheet

A dot is a joint. A line that hops over another does not touch it. Thick = bare rod, or one of the five wires that cross the two joints. Thin = resistor or capacitor leg.
Ground symbols: power grounds to the holder's − lead, signal grounds to the XIAO's GND pin. No 5 V rail exists.



Rev B, drawn 2026-11-10, verified on paper against the Seeed wiki, the MAX98357 datasheet and the vendor listings. Not built. The pin audit, the ladder audit and the line-clearance check in gen/wiring_board.py pass.

POCKET AI ASSISTANT · CAMERA BUILD · REV B

A tower that looks around, and up and down

The second Pocket AI: a XIAO ESP32-S3 Sense (camera + mic + charger) in a head that tilts inside a brass yoke and turns on the body, the owned screen, amp and speaker, two gear motors on one MX1508, and one Nitecore 16340 cell that charges through the XIAO's USB-C.
Rev B drawn 2026-11-10 for approval: adds the tilt axis, the yoke, and the three-button ladder. Rev A was 2026-10-27. Nothing built yet. Source of truth: `pocket_ai_assistant_with_camera/docs/WIRING.md` and `COMPONENTS.md`.

THREE RULES THAT DO NOT BEND

The cell is the only power source: nothing is ever fed into the XIAO's 5V pin, and nothing else is ever connected to its BAT pads. The speaker's two leads are not ground: no scope ground, no GND wire, ever. The brass frame is not a conductor: every joint is insulated from it, as in the first build.

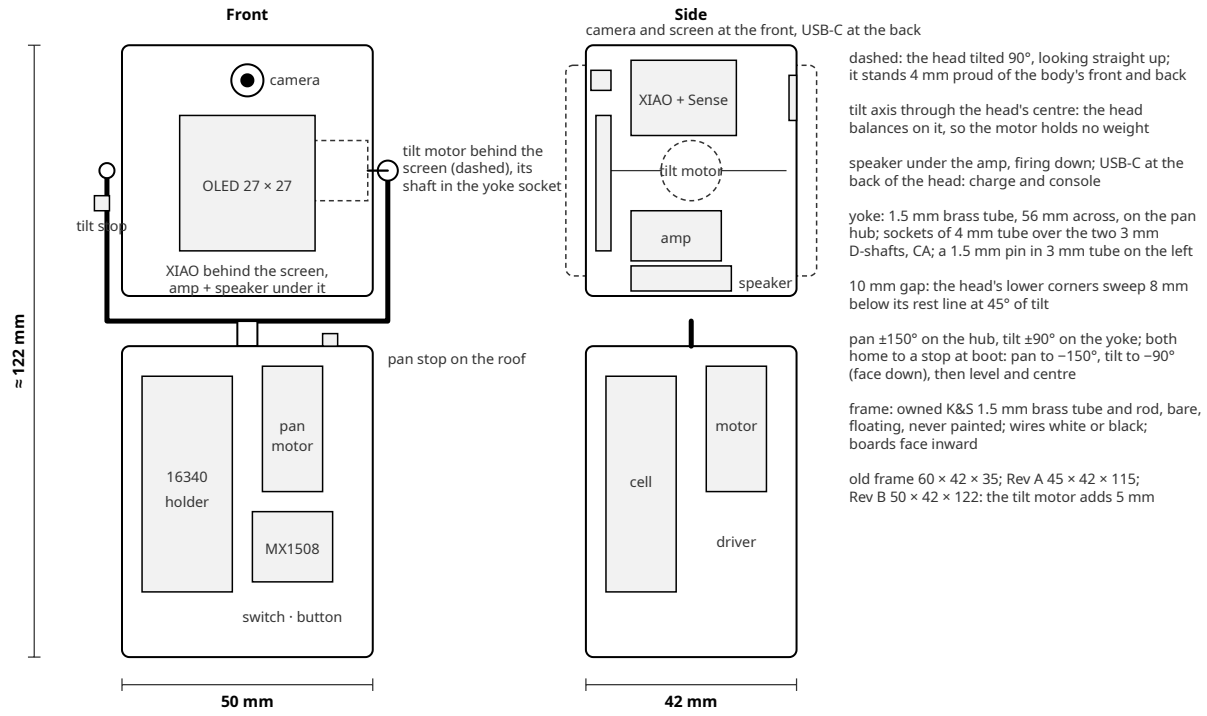
The pin map

All eleven header pins of the XIAO, all in use. Fixed by the Sense board: camera on GPIO 10-18, 38-40, 47, 48; PDM microphone on 41 (data) and 42 (clock); SD slot chip select on 21, SPI on 7/8/9.

HEADER	GPIO	NET	JOB
D0	1	BCLK	I2S bit clock to the amp
D1	2	LRC	I2S word clock to the amp
D2	3	DIN	I2S data to the amp
D3	4	AMP_SD	amp shutdown, pull-down R1 keeps it silent at boot
D4	5	SDA	I2C SDA, OLED
D5	6	IN3	tilt motor IN3 (LEDC PWM), pull-down R7
D6	43	SCL	I2C SCL, OLED. UART0 TX chatters here at boot: clock pulses with SDA held high are not a start condition, so the screen ignores them
D7	44	IN4	tilt motor IN4 (LEDC PWM), pull-down R8. UART0 RX is an input at boot; R8 holds the driver input low
D8	7	IN1	pan motor IN1 (LEDC PWM), pull-down R2
D9	8	IN2	pan motor IN2 (LEDC PWM), pull-down R3
D10	9	LADDER	the ladder: ADC1 channel 8 reads which of the action button, the pan stop and the tilt stop is pressed

The tower

The head carries everything that talks to the world: camera, microphone, screen, speaker, and the tilt motor whose shaft is fixed to the yoke. The yoke is a U of brass tube on the pan motor's shaft. The body carries everything heavy and still: the cell, the switch, the driver, the pan motor, the button.



Targets, not measurements: head 50 × 42 × 50 mm, yoke 56 mm across, body 50 × 42 × 62 mm, 122 mm overall, about 95 g with the cell. Pan ±150°, tilt ±90°. Runtime about 3 hours following and talking, 7 idle; charge 10-20 hours through the XIAO asleep, 2 hours in an external charger. Caliper every module, the motors most of all, before the frame is cut.

Why each wire

NET	FROM	TO	WIRE	WHY
3V3	XIAO 3V3	OLED VCC, R4	rod	The screen is the only load on the 3.3 V regulator; the XIAO itself and the camera take the rest of its 700 mA. R4 is the ladder's pull-up.
GND	holder − (the star)	XIAO BAT− pad, amp GND, driver GND, 220 µF; then OLED GND, the three ladder resistors, R1-R3, R7, R8 and C4 on the XIAO's GND pin	rod bus in the head and in the body; one 26 AWG wire across each joint; the holder's own lead to the star	One net, two solder points. The amp's and motors' return current goes straight to the holder's − lead, never through the XIAO's pads. Only the small signal grounds sit on the XIAO's GND header pin.
SDA	D4 (GPIO5)	OLED SDA	rod	Any S3 pin can be I2C through the GPIO matrix. Address 0x3C or 0x3D, as before.

SCL	D6 (GPIO43)	OLED SCL	rod	GPIO43 is UART0 TX and the boot ROM prints on it for a fraction of a second. An I2C start needs SDA to fall while SCL is high; SDA sits high on its pull-up, so the SSD1306 never sees a frame. The console goes over USB, so nothing is lost.
BCLK	D0 (GPIO1)	amp BCLK	rod	I2S out only: the microphone is the PDM mic on the Sense board (GPIO41/42), so there is no shared duplex bus.
LRC	D1 (GPIO2)	amp LRC	rod	
DIN	D2 (GPIO3)	amp DIN	rod	GPIO3 is a strapping pin on the S3; it only matters if the JTAG eFuse is burned, which it is not on a stock XIAO.
AMP_SD	D3 (GPIO4)	amp SD, with R1 4.7 kΩ to GND	rod	Same rule as the first build: the amp stays off until the firmware raises SD, and the pull-down beats the breakout's 1 MΩ pull-up. The line stays in Rev B because the amp must be shut down before its clocks stop, else the output can sit at DC.
BAT+	holder + → switch → XIAO BAT+ pad	amp VIN, driver VCC, 220 µF	rod bus in the body and in the head; one 26 AWG wire across each joint; a short wire stub to the XIAO's BAT+ pad	3.0-4.2 V straight from the cell. There is no 5 V rail in this design: the XIAO's 5V pin is dead on battery, and every part here accepts cell voltage.
IN1	D8 (GPIO7)	driver IN1, with R2 10 kΩ to GND	rod in the head, 30 AWG across both joints	Two pins drive one motor: IN1 high + IN2 low turns one way, swapped turns the other, both low coasts, PWM on the high one sets speed. Bridge A is the pan motor in the body.
IN2	D9 (GPIO8)	driver IN2, with R3 10 kΩ to GND	rod in the head, 30 AWG across both joints	R2/R3 hold both inputs low while the S3 boots and its pins float; drop them only if the meter shows the module has its own pull-downs.
IN3	D5 (GPIO6)	driver IN3, with R7 10 kΩ to GND	rod in the head, 30 AWG across both joints	Bridge B is the tilt motor. Same rules as IN1/IN2.
IN4	D7 (GPIO44)	driver IN4, with R8 10 kΩ to GND	rod in the head, 30 AWG across both joints	GPIO44 is UART0 RX, an input at boot with a weak pull-up inside the chip; against R8 that is 0.6 V at the driver, well under its 1.8 V threshold, so the tilt motor cannot twitch at boot.
M1	driver OUT A1 / A2	pan motor leads	the motor's own leads, in the body	Polarity does not matter: swap the two if 'left' turns right.
M2	driver OUT B1 / B2	tilt motor leads	the motor's own leads to the body: 30 AWG across both joints	The tilt motor lives in the head, the driver in the body. 0.36 A stalled over 150 mm of 30 AWG drops 20 mV: nothing.

LADDER	D10 (GPIO9, ADC1_CH8)	R4 10 kΩ to 3V3; C4 100 nF to GND; action button + 680 Ω, pan stop + 2.2 kΩ, tilt stop + 5.6 kΩ, each to GND	rod in the head, 30 AWG across both joints, rod in the body	Twelve signals, eleven pins: the three switches share one ADC pin. Nothing pressed reads 3.3 V; each switch pulls the pin to its own voltage (table below) and the pairs are distinct too, so the firmware never mistakes a stop for the button. C4 and the 10 kΩ source make a 1 ms filter against motor noise.
SPK	amp OUT+ / OUT−	speaker	its own 32 AWG leads	Neither lead is ground. Never clip a scope ground to either.

ONE CELL, NO BOOST

Every part on this sheet accepts 3.0-4.2 V: the XIAO has its own regulator and charger, the MX1508 runs from 2 V, the MAX98357A from 2.5 V, the OLED sits on the XIAO's 3.3 V. So the design has one rail, the cell itself. The PowerBoost, the 5 V harness points and the load-sharing worry of the first build all go away.

EYES FIRST, THEN THE HEAD

The camera sees about 65° across; the eyes drawn on the 0.96 inch screen can only look about 20° off centre before they run out of glass. So the firmware finds the face in the frame and slides the eyes toward it first. When the face drifts past the eyes' reach, the head turns on its motors until the face is centred again and the eyes settle back to the middle. Eyes jump, head follows a beat later: a saccade, the thing that makes a face read as alive. The camera and the screen share one head, so where the camera looks is where the eyes look, by construction; the difference you would otherwise build with a second gimbal is drawn in pixels.

TWO AXES, TWO GEAR MOTORS, ONE DRIVER

Pan from the body, tilt inside the head. The MX1508 has two bridges, so the second motor costs two pins and two pull-downs, not a board. The tilt axis runs through the head's centre of mass, so the tilt motor holds no weight and the gearbox's own friction keeps the head where it was left. Both motors were on hand on 2026-11-03.

WHY THE MX1508 AND NOT THE L298N

The L298N's bipolar outputs lose 2-3 V and its onboard regulator wants 7 V or more. From a single 16340 a real L298N would hand the motor about 1 V. The board the builder linked is the MX1508: MOSFET outputs, 2-10 V in, inputs that read 3.3 V as high. Four control pins, two bridges, and both are used now.

WHY A DC GEAR MOTOR NEEDS A HOME STOP

A DC motor has no idea where it is. The 16 rpm gearbox makes timed moves repeatable (100 ms is about 10°), but drift adds up, so the firmware drives each axis to its stop at boot and whenever it is unsure: tilt first, face down, then pan to the left stop, then up and centre. One switch per axis. A hobby servo would do all of this in one three-wire part; it wants 5 V and is louder and jerkier, so the motors stand.

TWELVE SIGNALS, ELEVEN PINS: THE LADDER

The header has eleven pins. Three I2S, one amp shutdown, two I2C, four motor inputs: ten. The action button, the pan stop and the tilt stop would make thirteen, so the three share one ADC pin through a resistor ladder: a 10 kΩ pull-up to 3.3 V, and each switch pulls the pin down through its own resistor. Nothing pressed is 3.3 V; the button alone 0.21 V, the pan stop 0.60 V, the tilt stop 1.18 V, both stops 0.45 V, and any combination with the button under 0.25 V. The firmware reads a band, not a level, and ignores the button while it is homing. 5 % resistors and the ADC's own error move each reading by 50 mV; the gaps are 200 mV or more. The script asserts that.

WHY SCL SITS ON THE UART PIN

GPIO43 is UART0 TX, and the boot ROM prints on it for about a hundred milliseconds before the firmware runs. Rev A left it spare for that reason. Rev B needs the pin, so it carries the one signal that boot chatter cannot disturb: the I2C clock. An I2C start is SDA falling while SCL is high; with SDA idle on its pull-up, clock pulses alone are nothing to the SSD1306. Any S3 pin can be I2C through the GPIO matrix, and the console goes over USB, so nothing is lost. GPIO44 (UART0 RX) is an input at boot and drives the tilt motor's IN4 through its pull-down, which holds the driver input at 0.6 V, far under its threshold.

WHY THE AMP KEEPS ITS SHUTDOWN LINE

An easy way to find a pin would be to leave the amp's SD open and let it run whenever the cell is on. The MAX98357's maker says to shut the amp down before stopping its clocks, else the output can sit at DC across the speaker. So R1 and the SD line stay, the amp is off until the firmware raises SD, and the pin came from the ladder instead.

WHY THE AMP RUNS FROM BAT+ AND THE OLED FROM 3V3

The XIAO's 3.3 V regulator is good for 700 mA, and the S3 with the camera can take 370 mA of that on its own. The amp (250 mA peaks) and a motor (360 mA stalled) would push it over, so they take the cell directly, which both are rated for. The OLED's 20 mA rides on 3V3 as before.

CHARGING THROUGH THE XIAO, AND ITS LIMIT

The XIAO's charger is an SGM40567 set to 50-100 mA (the wiki says 100, the datasheet 50, owners report both). The device awake draws more than that, so with USB in and the switch ON the cell still goes down unless the firmware deep-sleeps. In-device charging is therefore an overnight trickle with the device asleep: 10-20 hours from empty. The charger has no thermistor input: room temperature only, first cycle watched. The practical path is to pull the cell and use an external 16340 charger (about 2 hours); the protected NL169 is made for that, and thirteen cells means one is always charged. That charger is the one optional item not yet owned.

USB IN WITH THE SWITCH OFF

The XIAO runs from USB for flashing and bench work with no cell. The amp stays off (R1 holds SD low) and the driver idles at under 1 mA, so the dead BAT+ rail is harmless. Never run a motor or the amp in this state: the rail has no cell behind it.

THE 5V PIN

On the XIAO it is the USB VBUS, nothing more. On battery it reads 0 V. Nothing in this design touches it, and nothing ever feeds it.

BARE RODS WHERE NOTHING MOVES, FIVE WIRES WHERE IT DOES

The builder wires with bare rod, as the first harness did: the owned 1.0 mm brass, soldered to the header pins at height levels so no two nets touch. Inside the head and inside the body every connection on the sheet is a rod. The two joints are different: a rod cannot flex ten thousand times, so the five connections that cross them (BAT+ and GND down to the head in 26 AWG; the ladder and the tilt motor's pair in 30 AWG) are silicone wire, run up one yoke upright as a single line, with a loose coil round the pan hub and a loop at the tilt pin. The holder, the motors and the speaker keep their factory leads, and the XIAO's BAT pads on the back of the board get a 10 mm flexible stub each rather than a rod soldered to a surface pad. Resistance is not a concern: 50 mm of 1.0 mm brass is 4 mΩ, nothing at 1 A. The frame stays floating, so every rod keeps 2 mm from it, as before.

COLOUR, UNDER THE RULES OF 2026-11-07 AND 11-09

White, ivory or black as much as possible; the frame its own metal; no paint and no skins. The frame and the yoke are bare brass. The screen is white on black glass, the XIAO and its Sense board are black, the speaker, the holder and the switch body are black, the switch lever and the motors are bare metal, the wires that show

are white or black from the owned kits, the heat-shrink is white. The parts that come in other colours are placed where they do not show: the driver (red or blue) behind the holder in the body, the amp under the speaker, the OLED's board (blue on some packs) behind its own glass, the cell's printed wrap inside the holder. The one owned part this build does not use for colour reasons is nothing: there is no part to replace.

WHAT THE SENSE BOARD OWNS

Fourteen GPIOs for the camera, two for the PDM microphone, and the SD slot's SPI on GPIO7/8/9, which are the header's D8-D10. Those three carry the pan motor and the ladder here, so the SD slot stays empty for good. That leaves exactly the eleven header pins on the sheet, all eleven in use.

The ladder, state by state

10 kΩ pull-up to 3.3 V on D10; each switch to GND through its own resistor. Volts at the pin for every combination, nominal; the firmware reads bands halfway between neighbours and ignores the button while it homes.

PRESSED	VOLTS AT D10	FIRMWARE READS
none	3.30 V	nothing
action button	0.21 V	button
pan stop	0.60 V	pan stop
tilt stop	1.18 V	tilt stop
pan stop + tilt stop	0.45 V	pan stop + tilt stop
action button + pan stop	0.16 V	button
action button + tilt stop	0.19 V	button
action button + pan stop + tilt stop	0.15 V	button

Does the cell carry it?

The NL169 is a protected 950 mAh cell rated 2 A continuous. The device never asks for half of that.

LOAD	FED FROM	TYPICAL	PEAK	FIGURE FROM
XIAO ESP32-S3 Sense, Wi-Fi + camera streaming	3V3 regulator (inside the XIAO)	155 mA	366 mA	Seeed wiki, webcam example on battery
XIAO, Wi-Fi connected, camera idle	3V3 regulator	110 mA	150 mA	Seeed wiki, Wi-Fi active
OLED, all pixels	3V3 pin	20 mA	25 mA	SSD1306 module, typical
Amp + speaker at the cap of 65	BAT+ rail	60 mA	250 mA	0.47 W into 8 Ω at 4.2 V, 85 % efficient
Pan motor, running / stalled	BAT+ rail	100 mA	360 mA	12 mm class; this listing publishes none: measure yours

Tilt motor, running / stalled	BAT+ rail	100 mA	360 mA	same motor; the firmware moves one axis at a time

3V3 regulator	Camera streaming 366 + OLED 25 = about 390 mA peak, inside the regulator's 700 mA. The amp and the motors are deliberately not on it.
Cell, worst second	390 + 250 + 360 = about 1.0 A if a motor stalls while it talks and streams. The firmware moves one axis at a time; even both stalled at once would be 1.36 A, under the cell's 2 A trip. The switch and the 26 AWG leads see this current; the 30 AWG signal wires never do.
Cell, average	Listening and talking with the camera on and the head following: about 270 mA, so about 3 hours. Wi-Fi idle with the camera off: about 130 mA, about 7 hours. Deep sleep on the Sense board is 3 mA.
Charge	50-100 mA from the XIAO's USB-C at the back of the head: 10 to 20 hours from empty, and only with the device asleep. An external 16340 charger (optional, not owned) does it in about two.

Meter checks before the first power-up

- [ ] **Holder polarity.** Cell in, nothing else connected: meter DC V on the holder leads. Red lead must read +3.6 to +4.2 V. If it reads negative the leads are swapped; swap them at the holder, never at the pads. On the XIAO's back, the BAT− pad is the one nearer the USB-C socket.
- [ ] **No shorts, unpowered.** Switch OFF, cell out. Ω between the BAT+ rail and GND: open or above 1 kΩ (the 220 µF makes it climb). 3V3 to GND: above 1 kΩ. BAT+ to 3V3: open.
- [ ] **OLED silkscreen.** Hold the module next to the sheet and read the four pin names before you solder. Two vendors, two orders.
- [ ] **Driver inputs.** Unpowered, Ω from each of IN1-IN4 to GND on the bare module. Under 100 kΩ means it has its own pull-downs and R2, R3, R7, R8 can go; open means keep them.
- [ ] **The ladder, unpowered.** Ω from the S pin to GND with nothing pressed: open. Press the button: 680 Ω. The pan stop: 2.2 kΩ. The tilt stop: 5.6 kΩ. Both stops: 1.6 kΩ. If a reading is off by more than 10 % a resistor is in the wrong place.
- [ ] **Bench supply first.** Holder empty. SKY TOPPOWER at 3.8 V, limit 0.5 A, + to the switch's 'in', − to GND. Switch ON: the XIAO boots, 3V3 pin reads 3.25-3.35 V, draw under 200 mA with the screen on. Then the camera stream: under 400 mA, no resets.
- [ ] **The ladder, powered.** Meter DC V on D10: 3.3 V with nothing pressed, about 0.21 V on the button, 0.60 V on the pan stop, 1.18 V on the tilt stop, 0.45 V with both stops. The firmware's bands sit halfway between.

- [ ] **Motors, head off the shafts.** Raise the limit to 1.5 A. Each motor in turn: 50 % PWM one way, then the other. Note the running current; stall it with a finger for one second and note the peak. Under 0.5 A stalled is the pass. Then with the head on: tilt down to the stop, pan to the stop, and watch that the wires coil and uncoil without pulling.
- [ ] **First charge, watched.** Cell in, switch ON, USB in, device asleep. Meter in series with the cell's + lead on the mA range: expect roughly 100 mA (Seeed says 100; one datasheet says 50). Cell cooler than 45 °C by touch after twenty minutes. The red LED goes out when it is full; the first full charge is not left unattended.
- [ ] **Speaker.** Volume cap 65 in the firmware, GAIN pin open. Scope the speaker differentially (A−B, both grounds on GND, never on a speaker lead): under 2.7 V peak.

What changes in the firmware

The Pip character firmware (ESP-IDF 6.0.2) was written for the ESP32-C3. The port is a pin map and four drivers, not a rewrite:

- **Target** esp32s3; keep the volume cap of 65, the amp-off-until-SD rule and the 8.5 dBm transmit cap as a config option (the XIAO has a real antenna; the cap can probably be lifted after an A/B test). Console on USB-serial-JTAG, not UART0.
- **Audio out** I2S standard TX on GPIO1/2/3, amp SD on GPIO4, exactly as the sheet. SD low before the clocks stop, every time.
- **Microphone** I2S PDM RX on GPIO42 (clock) and GPIO41 (data) replaces the INMP441 driver. 16 kHz mono as before.
- **Screen** I2C on GPIO5 (SDA) and GPIO43 (SCL) through the GPIO matrix.
- **Camera** the esp32-camera component with the XIAO ESP32S3 Sense pin table; OV2640 and OV5640 both initialise from it. Frames go into PSRAM; a face box from the frame drives the eyes and, past ±20°, the head.
- **Motors** LEDC PWM on GPIO7/8 (pan) and GPIO6/44 (tilt) at 20 kHz (above hearing, under the MX1508's 50 kHz), duty capped at 70 %, one axis at a time. Home at boot: tilt down to its stop, pan to its stop, then level and centre; timed moves from there; re-home on any doubt.
- **Ladder** ADC1 channel 8 (GPIO9), 12 dB attenuation, read at 100 Hz with a 4-sample average; bands at 0.33, 0.52, 0.85 and 1.6 V; the button is ignored while homing. Short press talks, long press resets Wi-Fi.

None of this exists yet. The first bench step is the XIAO alone on USB with the screen, the same way the first build started.

Parts, with the exact listing

Nothing is left to buy as of 2026-11-10: the second motor, the stops and the ladder resistors are all on hand. The five Amazon listings are the ones the builder ordered on 2026-10-27; owned items are listed with the inventory that records them. The one part not yet in the building is the controller pair.

BOUGHT FOR THIS PROJECT, 2026-10-27 (AMAZON ORDER; PRICES NOT RECORDED)

QTY	PART	WHERE	PRICE	WHY, AND WHAT TO CHECK

2	Seeed XIAO ESP32-S3 Sense, with camera and microphone	Amazon B0C69FFVHH	The brain, the camera and the microphone in one 21 × 18 mm stack: ESP32-S3, two cores at 240 MHz, 8 MB PSRAM (the camera needs it), USB-C, a Li-ion charger on the back, a 700 mA 3.3 V regulator, and the Sense expansion board with the camera (OV2640 or OV5640, same pins, about 65° wide) plus a PDM mic. Black boards, silver shield. The pair (order 114-1888127-6908215) has not shipped as of 2026-11-10; Amazon's estimate is 11-10 to 11-15. The second is the Pocket AI Sphere's, not a spare. On arrival: note which sensor is printed on the camera module, and keep the SD slot empty.
1 pack of 5	MX1508 dual H-bridge driver, Aideepen	Amazon B075S368Y2	Sold as an L298N replacement, but it is the MOSFET MX1508: 2-10 V motor supply, 1.5 A continuous per bridge, inputs happy at 3.3 V, no logic-supply pin. Both bridges are used now: A pans, B tilts. The board comes red or blue; it sits in the body behind the holder, out of sight. On arrival: the board should say MX1508 and carry VCC, GND, IN1-IN4 and two motor outputs. Delivered 2026-10-30.
2 (both on hand 2026-11-03)	Micro gear motor 1218GEN20, 3 V 16 rpm, Naroote	Amazon B0BS9K8Q8Z	12 mm metal gearmotors, 3 mm D-shaft, 16 rpm at 3 V. One pans from the body, one tilts from inside the head. Slow is what a head wants: at the cell's 3.7-4.2 V run them at 70 % PWM or less, one axis at a time. Stall current is not published; the 12 mm class is about 0.36 A, so measure it on the bench supply before either goes in the frame. Caliper the body length and the shaft before the yoke is cut.
1 pack of 10	16340 / CR123A holder with leads, ACEIRMC	Amazon B0CRGK889F	The holder from the old fit check (43 × 18 × 14 mm), black. CR123A and 16340 are the same can; the NL169 is 16.6 × 34.1 mm with a button top. On arrival: check that the spring reaches the button and meter that red is on the + contact. Delivered 2026-10-30.

1 pack of 20	MTS-102 mini toggle switch, Ferwooh	Amazon B0CZ6XZ2P9		SPDT ON-ON, 3 solder lugs, 6 A at 125 V AC: far above the 1 A peak it carries in the cell's + lead. Use the middle lug and one outer lug; the third stays empty; heat-shrink both (white). The owned Chanzon slide switches are bench-only here: not rated for that current. Delivered 2026-10-30.

OWNED, IN POCKET_AI_ASSISTANT/DOCS/INVENTORY.MD: DO NOT BUY

QTY	PART	WHERE	PRICE	WHY, AND WHAT TO CHECK
1	Hosyond SSD1306 0.96" OLED	owned ×10		Same screen, same I2C, same silkscreen trap (GND-VCC-SCL-SDA or VCC-GND-SCL-SDA: read it). White pixels on black glass; the board behind the glass is blue on some packs, so check the drawer and face it inward.
1	HiLetgo MAX98357A amplifier	owned ×12		Runs from the cell (2.5-5.5 V in). GAIN left open = 9 dB. Volume cap stays at 65: at a full 4.2 V cell that is 0.47 W into the 0.8 W speaker. Under the speaker in the head, board side in.
1	Same Sky CES-20134-088PM speaker	owned ×6		Black. Its 60 mm leads reach from the amp in the head without any wire of yours.
3	QTEATAK tactile buttons	owned ×420		The action button, the pan stop and the tilt stop. White caps on the action button.
9	Resistors: 680 Ω, 2.2 kΩ, 4.7 kΩ, 5.6 kΩ, 10 kΩ ×5	LuminologyPro kit		R1 amp SD pull-down (4.7 kΩ); R2, R3, R7, R8 driver input pull-downs (10 kΩ); R4 ladder pull-up (10 kΩ); the ladder's 680 Ω (button), 2.2 kΩ (pan stop), 5.6 kΩ (tilt stop).
4	Capacitors: 220 µF, 10 µF, 100 nF ×2	ALLECIN and BOJACK kits		220 µF on the BAT+ rail at the amp; 10 µF + 100 nF at the OLED's VCC pin; C4 100 nF at D10 for the ladder.
	26 AWG (power) and 30 AWG (signal) silicone wire, headers, heat-shrink, Kapton	TUOFENG, CBAZY, Pointool, ELEGOO, XFJYMXDM		26 AWG for BAT+ and GND across the joints, 30 AWG for the other three. White or black wire only where it shows: the five run up one yoke upright as one deliberate line. White heat-shrink.
	K&S 1.5 mm tube/rod, 1.0 mm rod	owned		Frame stock for the tower and the yoke: brass tube for the outline and the yoke, 1.0 mm rod for the harness, same soldering as the first frame. Bare brass, no paint (builder's rule 2026-11-09). The frame stays electrically floating.

OWNED, IN CYBER_CLOCK_V2/DOCS/INVENTORY.MD

QTY	PART	WHERE	PRICE	WHY, AND WHAT TO CHECK
1 (of 13)	Nitecore NL169 16340, 3.6 V 950 mAh, protected button-top	owned ×13 (IMR Batteries, delivered 2026-10-21)		The cell. Protected (2 A continuous, 4.20 V termination), so the XIAO's plain 4.2 V charger suits it. Never solder to it; it sits in the holder, in the body, where its printed wrap does not show.

2 + 1	M METERXITY brass tube set, 2 to 7 mm	owned (three sets, delivered 2026-10-28)		Two sockets from the 4 mm × 0.5 mm wall tube (3 mm bore) over the two 3 mm D-shafts: the pan hub on the yoke's base and the tilt socket on the yoke's right upright; a drop of CA or a flat filed on the shaft locks each. One short piece of the 3 mm tube on the left upright is the bearing for the head's 1.5 mm pivot pin. Caliper all of it before cutting.

OPTIONAL, NOT OWNED

QTY	PART	WHERE	PRICE	WHY, AND WHAT TO CHECK
1	External 16340 / CR123A Li-ion charger (Nitecore UI1/UI2 or XTAR VC2)	any, about $10-15		The fast way to charge: pull the cell, two hours, done. The XIAO's own charger is a 50-100 mA trickle for an asleep device. Not required to build or run.

NOT NEEDED ANY MORE

QTY	PART	WHERE	PRICE	WHY, AND WHAT TO CHECK
	ESP32-C3 SuperMini, INMP441 microphone, PowerBoost 1000C copy, fish paper	stay in the drawer		The XIAO carries the mic and the charger, nothing here needs 5 V, and a hard-can cell in a plastic holder needs no jacket.

Built by `docs/pages/gen/wiring_board.py`. The script audits the pin map against the Sense board's reserved GPIOs, audits the ladder's voltages against 5 % resistors, and checks every drawn line for clearance before it writes anything. The electrical figures come from the Seeed wiki, the MAX98357 datasheet and vendor listings, not from measurement.